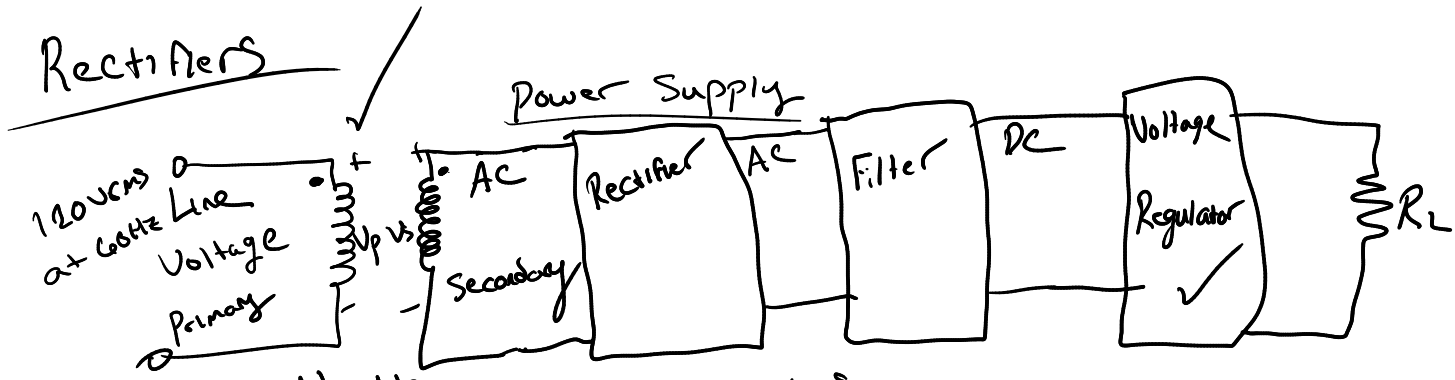
Rectifiers



$N_p : N_s$

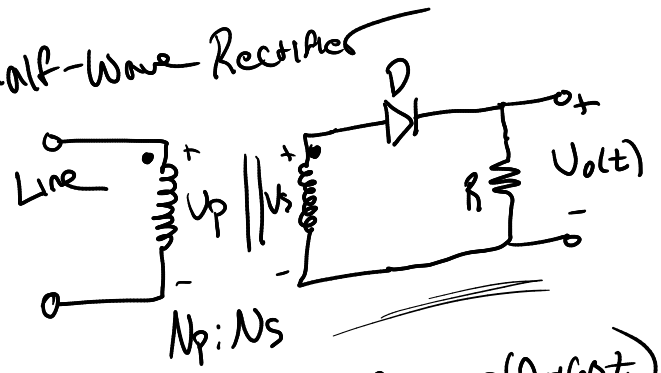
$$\frac{V_p}{V_s} = \boxed{\frac{N_p}{N_s}}$$ turns ratio

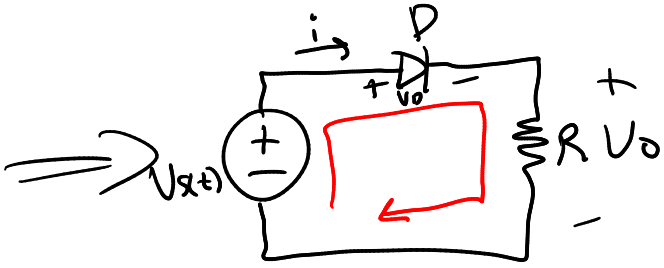
10 Vrms → 5 Vrms
Vp Vs
2 : 1

$V_s = V_m \cos(2\pi f t)$
or V_{peak}

$\frac{10}{5} = 2$

Half-Wave Rectifier





KVL: $-V_s(t) + V_D + iR = 0$

This equation holds no matter if D is ON or OFF

$V_o = iR$

We want $V_s(t) = 200\cos(2\pi 60 t)$

When $V_s(t) < 0$ D is OFF

$i = 0$ $V_o = 0$

$-V_s(t) + V_D + \cancel{iR}^{\,0} = 0$

$V_D = V_s(t)$

When $V_s(t) > 0$ D is ON

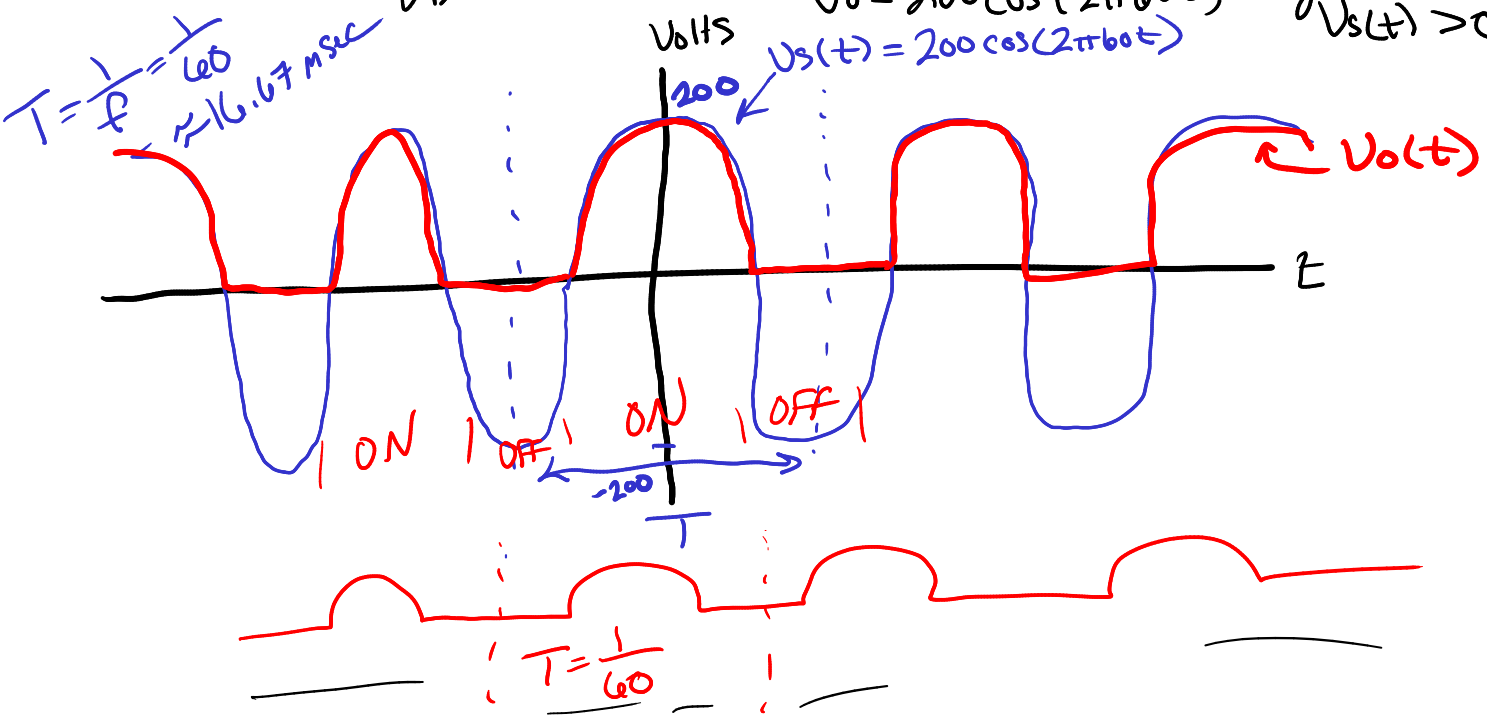
$V_D = 0$

$-V_s(t) + \cancel{V_D}^{\,0} + V_o = 0$

$V_o(t) = V_s(t)$

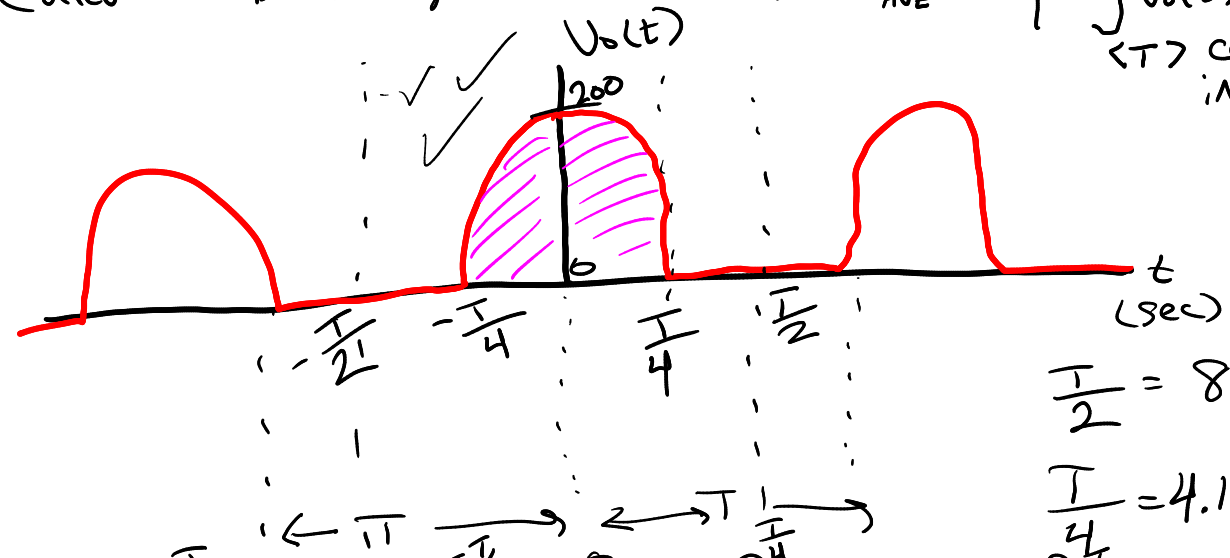
$V_o = 200\cos(2\pi 60 t)$ only when $V_s(t) > 0$

$T = \frac{1}{f} = \frac{1}{60} \approx 16.67$ msec



Calculate DC average of $V_o(t) \Rightarrow V_{o_{AVE}} = \frac{1}{T}\int V_o(t)\,dt$

<T> contiguous interval T sec long



$\frac{T}{2} = 8.33$ msec

$\frac{T}{4} = 4.1667$ msec

$$V_{o_{AVE}} = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} V_o(t)\,dt = \cancel{\frac{1}{T}\int_{-T/2}^{-T/4} V_o(t)\,dt}^{\,0} + \frac{1}{T}\int_{-\frac{T}{4}}^{\frac{T}{4}} V_o(t)\,dt + \cancel{\frac{1}{T}\int_{\frac{T}{4}}^{\frac{T}{2}} V_o(t)\,dt}^{\,0}$$

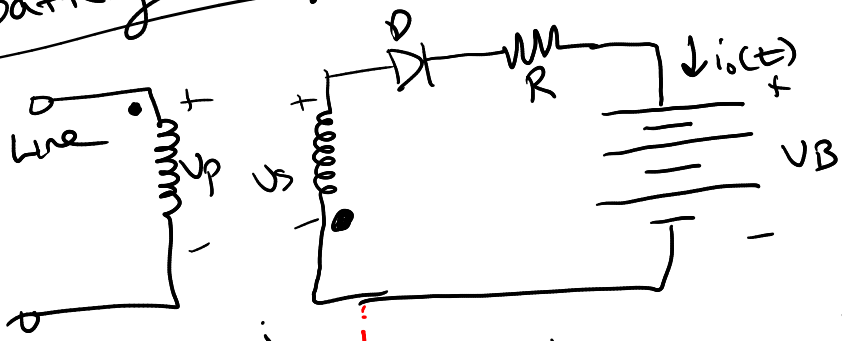
$$V_{o_{AVE}} = \frac{1}{T}\int_{-\frac{T}{4}}^{\frac{T}{4}} 200\cos(2\pi 60t)\,dt = \frac{2}{T}(200)\int_0^{\frac{T}{4}} \cos(2\pi 60t)\,dt$$

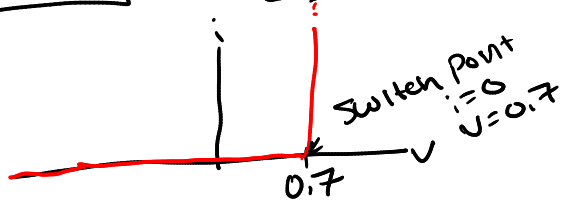
$$\frac{1}{T} = f = 60 \quad = \frac{2}{T}(200)\left[\frac{1}{2\pi 60}\sin(2\pi 60t)\right]_{t\to 0}^{t\to \frac{T}{4}}$$

$$V_{o_{AVE}} = \frac{(60)(400)}{2\pi 60}\sin\left(2\pi 60\,(4.1667\times 10^{-3})\right)$$

$$\boxed{V_{o_{AVE}} = 63.662 \text{ Volts}}$$

Battery Charger





Given $V_B = 1.5$ Volts

Line: 120 Vrms @ 60 Hz

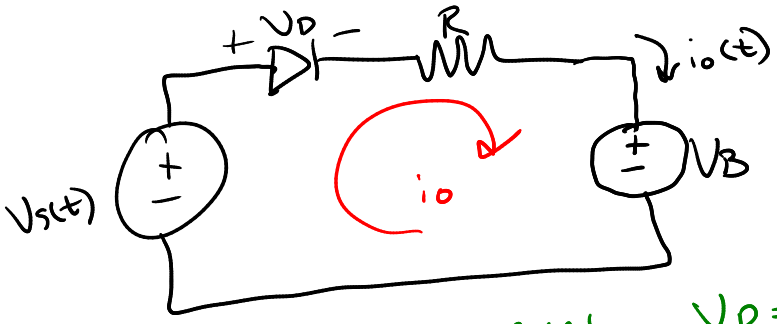
Suppose $V_{s_{peak}} = 3.0$ Volts

$V_s(t) = 3\cos(2\pi 60t)$ Volts

CVD7 model

w/ $V_{zk} \to \infty$ volts

$r_z = 0\,\Omega$



KVL: $-V_s(t) + V_D + R i_o(t) + V_B = 0$

Holds under all conditions

@ $t = t_1$ @ switching point ... $V_D = 0.7$, $i_o(t) = 0$

Make sure you are in radians

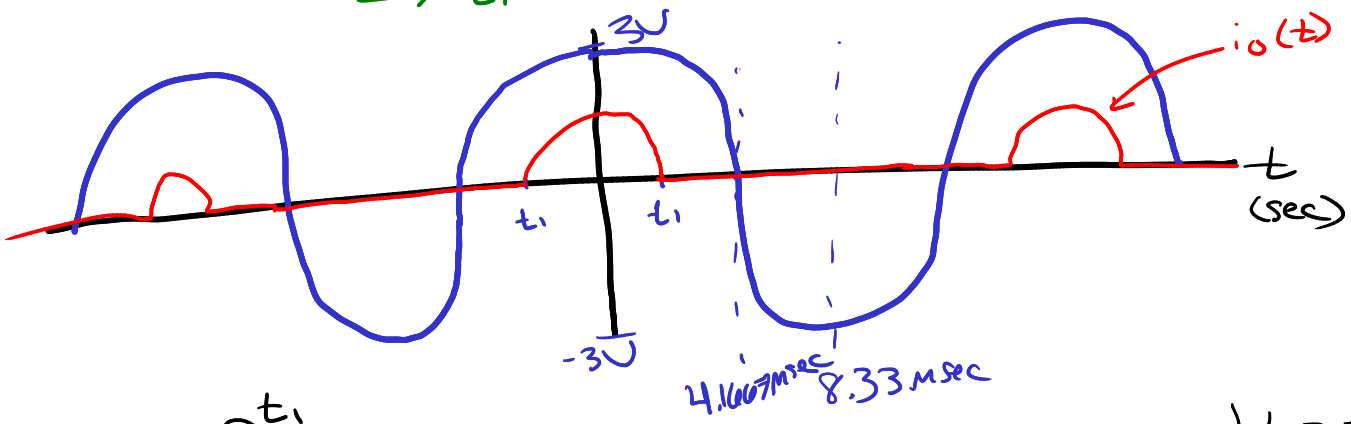
$$-V_s(t_1) + 0.7 + 0 + 1.5 = 0$$

$$\Rightarrow V_s(t_1) = 2.2 = 3\cos(2\pi 60 t_1)$$

$$\Rightarrow \cos(2\pi 60 t_1) = \frac{2.2}{3.0} \Rightarrow \cos^{-1}\left(\frac{2.2}{3.0}\right) = 2\pi 60 t_1$$

$$\Rightarrow t_1 \approx 1.983 \text{ msec}$$

$T = \frac{1}{60} \approx 16.67$ msec



$$I_{o_{AVE}} = \frac{2}{T}\int_0^{t_1} i_o(t)\,dt \qquad i_o(t) > 0 \text{ between } t=0 \text{ and } t = t_1$$

KVL: $-V_s(t) + V_D + R i_o + V_B = 0$

$$-3\cos(2\pi 60 t) + 0.7 + 1.5 = -R\, i_o(t)$$

$$i_o(t) = \frac{1}{R}\left[3\cos(2\pi 60 t) - 2.2\right]$$

$$I_{o_{ave}} = \frac{2}{T}\frac{1}{R}\int_0^{t_1}\left[3\cos(2\pi 60 t) - 2.2\right]dt$$

$$= (60)(2)\frac{1}{R}\left[\frac{3}{2\pi 60}\sin(2\pi 60 t) - 1.2t\right]_{t=0}^{t=t_1}$$

if we let $R = 1k\Omega$

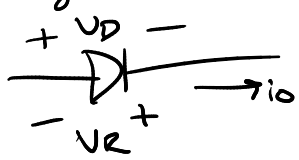
$$I_{o_{AVE}} \cong 126\,\mu\text{Amps}$$

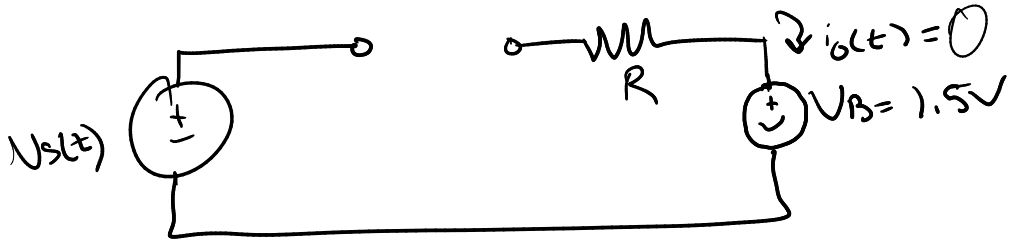
we assumed $V_{ZK} \to \infty$ Volts

In Reality V_{ZK} is finite

Find Peak Inverse Voltage : V_{PIV}

assume D is OFF





KVL: $-V_S(t) - V_R(t) + 0 + 1.5 = 0$

$V_R(t) = 1.5 - V_S(t) = 1.5 - 3\underbrace{\cos(2\pi 60 t)}$

$V_{PIV} = \max(V_R(t))$ ... occurs when -1

$\boxed{V_{PIV} = 4.5 \text{ volts}}$

want $V_{ZK} \gg 4.5$ volts

maybe 10 volts